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Batch No:3IT15

20IT281 – Mini Project II

"PREDICTIVE CROWD MANAGEMENT AND RESOURCE ALLOCATION FOR PUBLIC SECTORS"

SDG GOALS MAPPED Nos. :09,11,16

Type of Project: Application / Society Based

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Zeroth Review
Date : 25.07.2026

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Introduction

- Existing government queue systems primarily use **physical/digital tokens and appointment scheduling** to manage citizens in a First-Come-First-Served manner.
- These systems provide basic queue information but have limited ability to **dynamically predict waiting time or anticipate crowd congestion**.
- Government service time varies depending on **service type, document verification, citizen arrival rate, queue length, and counter/officer availability**.
- The proposed system integrates **digital appointment booking, virtual queue management, and AI/ML-based waiting-time prediction** into a unified platform.
- The AI/ML model analyzes **real-time queue data and historical patterns** to predict waiting time and forecast potential queue congestion.
- Citizens receive **digital tokens, live queue positions, predicted waiting times, and turn notifications**, reducing unnecessary physical waiting.
- Administrators can monitor queues through a dashboard and receive **crowd forecasts, queue analytics, and intelligent counter-allocation recommendations**.

Literature Survey - 01

Experiment	Inference	Problem gap
Developed a machine-learning-based waiting time prediction system using historical queue data. The study formulates waiting-time prediction as a **regression problem** and evaluates ensemble regression models such as **Random Forest, XGBoost, Gradient Boosting, and Extra Trees** to estimate customer waiting time accurately.	The ensemble regression models achieved high prediction accuracy, with tree-based methods outperforming conventional regression techniques. The model provides reliable real-time waiting-time estimates, enabling better queue monitoring and service planning while reducing uncertainty for customers.	The proposed system mainly focuses on **waiting-time prediction** and does not include **dynamic resource allocation**, **automatic counter management**, **appointment scheduling**, or **government-office workflows**.commend opening additional service counters during peak crowd conditions.

- **Source:** T. K. Taton *et al.*, **"Waiting Time Prediction in Queue Management: Leveraging Machine Learning Approach,"** Proceedings of the 2024 IEEE International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET), 2024, DOI: **10.1109/ICRPSET64863.2024.10955915**

Literature Survey - 02

Experiment	Inference	Problem gap
Evaluated Random Forest, Extra Trees, Gradient Boosting, Histogram Gradient Boosting, Voting and Ridge Regression using historical bank queue data to predict numerical waiting time.	Tree-based ensemble ML models are effective for **queue waiting-time prediction**. The study found the **Extra Trees Regressor** to be the best-performing model among those evaluated.	The study mainly focuses on **prediction accuracyusing historical bank data**. It does not implement a complete government queue system with appointment booking, live citizen notifications and dynamic counter allocation.

- **Source:** Tapodhir Karmakar Taton, Bipin Saha, Md. Johirul Islam and Shaikh Khaled Mostaque, ***"A Comprehensive Approach to Queue Waiting Time Prediction using Tree-Based Ensembles with Data Balancing and Explainable AI,"*** **Discover Analytics**, Vol. 3, Article 9, 2025, DOI: 10.1007/s44257-025-00037-2.

Literature Survey - 03

Experiment	Inference	Problem gap
Developed a **machine learning-based patient waiting-time prediction** model using historical hospital data to estimate treatment time and reduce queue delays.	The predictive model improved waiting-time estimation and supported better queue planning and resource utilization.	The system is designed for **hospital queues** only. It does not support **government office workflows**, **dynamic counter allocation**, **appointment scheduling**, or **citizen token management**.

- **Source:** Mai Abusair, Mohammad Sharaf, Tuqa Hamad, Raghad Dahman *et al.*, **"Reducing Patient Waiting Time at Public Hospitals Through Predictive Modeling,"** *IEEE ZCICT 2024*, DOI: **10.1109/ZCICT63770.2024.10958517**.

Literature Survey - 04

Experiment	Inference	Problem gap
Developed a **machine-learning-based queue length detection system** using video surveillance data. The study formulates **queue length detection as a regression problem.**The system estimates the number of people waiting in a queue from camera frames to support efficient queue monitoring in public institutions.	The ensemble regression models achieved high prediction accuracy, with tree-based methods outperforming conventional regression techniques. The model provides reliable real-time waiting-time estimates, enabling better queue monitoring and service planning while reducing uncertainty for customers.	The proposed system mainly focuses on **waiting-time prediction** and does not include **dynamic resource allocation**, **automatic counter management**, **appointment scheduling**, or **government-office workflows**.owd conditions.

- **Source:** Mehmet Eren Yeşilyurt *et al.*, **"A Comparison of Machine Learning Methods for Queue Length Detection,"** *Communications Faculty of Sciences University of Ankara*, Vol. 66, No. 2, pp. 132–139, 2024. DOI: **10.33769/aupse.1415447**.

Literature Survey - 05

Experiment	Inference	Problem gap
Reviews existing queue-management studies and proposes an integrated framework combining queue-state data capture, feature engineering, ML waiting-time prediction, dynamic counter allocation and monitoring for public services.	Combining waiting-time prediction with dynamic counter allocation can support better resource utilization, congestion management and transparent public-service delivery.	The work primarily proposes a framework based on reviewed literature; extensive real-world implementation and cross-domain validation are still required.

- **Source:** Dr. Balamurugan S. and Dhanush S., "**ML-Based Smart Queue Management System for Public Services**," IRE Journals, Vol. 9, Issue 11, pp. 2662 onward, May 2026, ISSN: 2456-8880, DOI: 10.64388/IREV9I11-1717976

Problem identification, scope and objectives (Contd.,)

Problem Definition :

- Government offices face **long queues, overcrowding, and unpredictable waiting times**, especially during peak hours.
- Existing token-based systems mainly manage queue order but lack **dynamic waiting-time prediction and congestion forecasting**.
- Variations in **service time, queue length, citizen arrivals, and counter availability** make effective queue management difficult.
- Therefore, an **AI-based adaptive queue management system** is required for efficient and predictable public service delivery.

Scope:

- **Online appointment booking** and digital token generation.
- **Virtual queue** and real-time queue tracking.
- **AI/ML-based waiting-time prediction** using real-time and historical data. **Crowd forecasting** to identify peak periods and congestion.
- **Intelligent counter-allocation recommendations** for administrators.
- **Real-time notifications** and queue updates for citizens.
- Scalable to **multiple Tamil Nadu government offices and services**.

Objectives :

- **To develop an intelligent digital queue management system** that enables citizens to book appointments, generate digital tokens, and track their live queue position.
- **To implement an AI/ML-based prediction model** for dynamically estimating citizen waiting time and forecasting queue congestion using real-time and historical queue data.
- **To optimize government service operations** by providing intelligent counter-allocation recommendations, real-time queue updates, and turn notifications to reduce waiting time and overcrowding.

Existing technology of the project

- Existing queue management systems use **physical/electronic token generation** to maintain the order of customers or citizens.
- **Online appointment and virtual queue systems** allow users to reserve service slots and remotely monitor queue status.
- Traditional waiting-time estimation commonly depends on **queue length, average service time, and predefined/static rules**.
- Recent research applies **Machine Learning regression models** to historical and real-time queue information for more accurate waiting-time prediction.
- Ensemble ML techniques such as **Random Forest, Extra Trees, and Gradient Boosting** have been investigated for queue waiting-time prediction.
- Existing research demonstrates that ML can improve waiting-time estimation, but integrating **dynamic prediction, congestion forecasting, and resource/counter recommendations** into one government-service platform remains an important area for development.

Source: **Waiting Time Prediction in Queue Management: Leveraging Machine Learning Approach," T. K. Taton et al., IEEE, 2024/2025.** The study investigates machine-learning-based queue waiting-time prediction as a regression problem using ensemble methods

Existing technology of the project

Photographs:



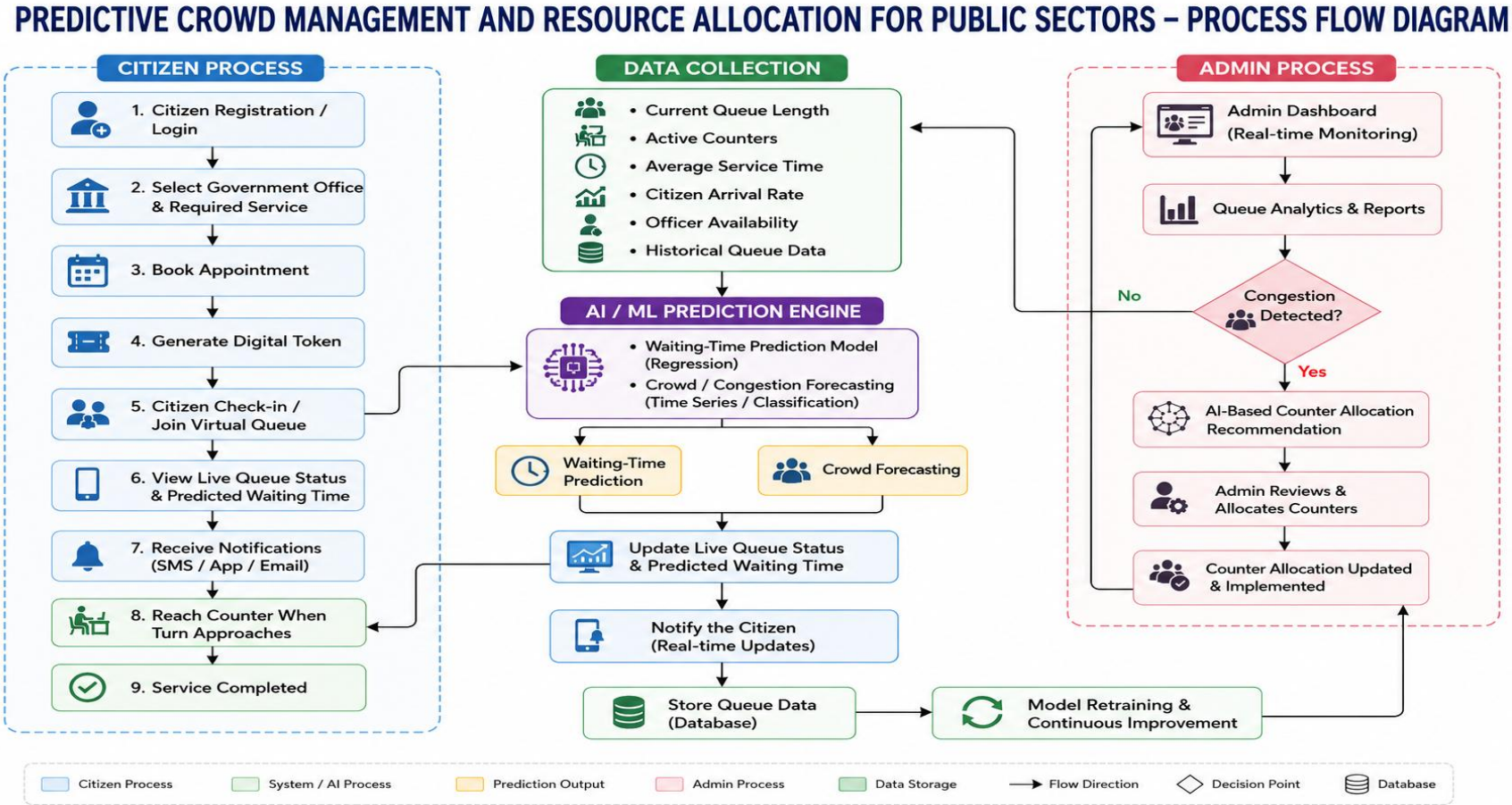




- Existing systems primarily rely on physical queues, digital tokens, appointment scheduling, and electronic queue displays for managing citizen services.

Methodology

Process Flow Diagram



Technical feasibility

- **Web Application Development:** React.js and Bootstrap provide a responsive and user-friendly interface for citizens, officers, and administrators.
- **Backend & Database:** Node.js, Express.js, and MongoDB support appointment booking, digital tokens, queue management, user data, and historical queue records.
- **AI/ML Implementation:** Python and Scikit-learn are used to develop models for waiting-time prediction and crowd forecasting using real-time and historical queue data.
- **Real-Time Queue Updates:** The system continuously updates queue position, current token, counter availability, and predicted waiting time.
- **Notification Service:** Firebase can provide appointment reminders, queue updates, and turn notifications to citizens.
- **Scalability:** The system can initially support selected government offices and can later be extended to multiple departments and locations.
- **Technology Stack**
- **Frontend:** React.js + Bootstrap
- **Backend:** Node.js + Express.js
- **Database:** MongoDB
- **AI/ML:** Python + Scikit-learn
- **ML Integration:** Flask / FastAPI

Time line

Month / Period	Phase	Key Action Items	Target Deliverable
July – Aug 2026	Phase I	• Problem identification • Requirement analysis • Literature survey and analysis • Finalize system modules	Requirements & Literature Review
Sept – Oct 2026	Phase I	• UI/UX design (citizen & admin) • System architecture design • Database design • Queue dataset collection & preparation	System Design & Dataset
Nov – Dec 2026	Phase I	• Develop citizen and admin modules • Implement appointment & digital token system • Develop Random Forest model for waiting-time prediction	Working Prototype & ML Model
Jan – Feb 2027	Phase II	• Integrate ML model with application • Implement real-time queue tracking • Implement notifications (SMS/Push) • Develop counter recommendation module	Integrated AI Queue System
Mar – Apr 2027	Phase II	• System testing (functional & performance) • ML model evaluation & optimization • User acceptance testing • Final documentation & project demo	Final System & Project Report

SDG Goals

- **SDG 9 – Industry, Innovation and Infrastructure**
- Promotes the use of AI/ML and digital technologies in government service infrastructure.
- Supports the modernization of traditional queue-management systems through intelligent automation and data-driven solutions.
- **SDG 11 – Sustainable Cities and Communities**
- Helps reduce overcrowding and unnecessary physical waiting at government offices.
- Supports smarter and more accessible public-service delivery through digital appointments and virtual queues.
- **SDG 16 – Peace, Justice and Strong Institutions**
- Improves the efficiency, accessibility, and transparency of government service delivery.
- Enables better queue monitoring and resource utilization through AI-based predictions and administrative decision support.

References

- **[1]** T. K. Taton *et al.*, "Waiting Time Prediction in Queue Management: Leveraging Machine Learning Approach," *Proceedings of the 2024 IEEE International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*, 2024, doi: 10.1109/ICRPSET64863.2024.10955915.
- [2] T. K. Taton, B. Saha, M. J. Islam and S. K. Mostaque, "A Comprehensive Approach to Queue Waiting Time Prediction Using Tree-Based Ensembles with Data Balancing and Explainable AI," Discover Analytics, vol. 3, article 9, 2025, doi: 10.1007/s44257-025-00037-2.
- M. Abusair, M. Sharaf, T. Hamad, R. Dahman *et al.*, "Reducing Patient Waiting Time at Public Hospitals Through Predictive Modeling," *IEEE ZCICT 2024*, 2024, doi: 10.1109/ZCICT63770.2024.10958517.
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- [5] S. Balamurugan and S. Dhanush, "ML-Based Smart Queue Management System for Public Services," IRE Journals, vol. 9, no. 11, May 2026, doi: 10.64388/IREV9I11-1717976